

Qualification Pack



Building Management System Service Engineer

QP Code: ELE/Q7104

Version: 3.0

NSQF Level: 5

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ELE/Q7104: Building Management System Service Engineer

Brief Job Description

A Building Management Systems (BMS) Service Engineer is responsible for installing the Building Management System (BMS) at the client premises. The person is also responsible for monitoring the BMS for the correct functioning and carrying out its regular repair and maintenance.

Personal Attributes

The individual must have attention to detail along with problem-solving skills. The person must be physically fit to work for long hours with concentration and have the ability to coordinate with others to achieve the work objectives. Good communication skills and strong organisational skills are other important attributes required in this job role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N7205: Prepare for installing the BMS](#)
2. [ELE/N7206: Carry out the installation of BMS](#)
3. [ELE/N7208: Carry out commissioning and testing of BMS](#)
4. [ELE/N7207: Carry out repair and maintenance of BMS](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service
Country	India
NSQF Level	5
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7411.0100

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Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (Physics/Electronics/Electrical/Mechanical) with 1.5 years of experience Relevant Exp in Industrial Automation OR Completed 3 year diploma after 10th (Electronics/Electrical/Mechanical) with 3 Years of experience Relevant Exp in Industrial Automation OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Relevant Exp in Industrial Automation
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-05-EH-03979-2025-V3-ESSCI
NQR Version	3.0

Remarks:

NA

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ELE/N7205: Prepare for installing the BMS

Description

This NOS unit is about to coordinate, verify, and prepare all necessary resources, personnel, and site conditions for the efficient and safe installation of Building Management System (BMS) components as per project requirements.

Scope

The scope covers the following :

- Check the availability of required resources
- Check and test the received resources
- Prepare for installing the BMS

Elements and Performance Criteria

Check the availability of required resources

To be competent, the user/individual on the job must be able to:

- PC1.** co-ordinate with the relevant company personnel/ third-party supplier to ensure timely delivery of all the necessary apparatus for the installation of BMS such as Direct Digital Controller (DDC), sensors, actuators, relevant types of cables, Human Machine Interface (HMI) display, computer, server, etc.
- PC2.** check that all the components are delivered as per the order
- PC3.** arrange the necessary tools, equipment, and Personal Protective Equipment (PPE) for the installation
- PC4.** ensure the availability of trained personnel to assist with the installation process

Check and test the received resources

To be competent, the user/individual on the job must be able to:

- PC5.** examine the BMS apparatus to ensure they are not physically damaged
- PC6.** test the BMS apparatus to ensure the correct functioning
- PC7.** report any issues identified with BMS apparatus promptly to the relevant personnel/ supplier
- PC8.** co-ordinate with the relevant personnel/ supplier to get a timely replacement of damaged/ faulty components
- PC9.** maintain the record of receipt/ replacement and testing of the apparatus

Prepare for installing the BMS

To be competent, the user/individual on the job must be able to:

- PC10.** check that all the civil works are completed in the building and necessary approval/ clearance is obtained before starting the installation process
- PC11.** prepare the circuit diagram/ shop drawing/ as-built drawings for the installation of BMS in the building
- PC12.** check that all the outlets related to BMS are accessible and not covered by Mechanical, Electrical and Plumbing (MEP) services
- PC13.** ensure the work area is ready and safe to start the installation of BMS systems

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- PC14.** plan the installation process to ensure timely completion as per the client requirements
- PC15.** co-ordinate with the Heating, Ventilation and Air-conditioning (HVAC) installation technician for the installation of HVAC equipment as per the BMS installation plan and client requirements
- PC16.** identify the locations for the installation of various BMS equipment/ devices such as access control, video surveillance, fire alarms, HVAC control, programmable lighting and electric power management as per the BMS installation plan
- PC17.** prepare various equipment for installation as per the manufacturer's instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the apparatus required for the installation of BMS such as Direct Digital Controller (DDC), sensors, actuators, relevant types of cables, Human Machine Interface (HMI) display, computer, server, etc.
- KU2.** the necessary tools, equipment, and Personal Protective Equipment (PPE) required for BMS installation
- KU3.** the importance of ensuring that trained personnel assist with the BMS installation process
- KU4.** the process of examining the BMS apparatus before installation to ensure no physical damage or malfunctions
- KU5.** applicable documentation requirements
- KU6.** the importance of ensuring all the civil works are completed in the building and necessary approval/ clearance is obtained before starting the installation process
- KU7.** how to prepare the circuit diagram/ shop drawing/ as-built drawings for the installation of BMS in the building
- KU8.** the importance of ensuring all the outlets related to BMS are accessible and not covered by Mechanical, Electrical and Plumbing (MEP) services
- KU9.** the importance of ensuring the work area is ready and safe to start the installation of BMS systems
- KU10.** the process of installation of HVAC equipment as per the BMS installation plan
- KU11.** the process of assembling various BMS equipment such as access control, video surveillance, fire alarms, HVAC control, programmable lighting and electric power management for installation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write work-related notes and maintain records
- GS2.** read the relevant literature to keep abreast with the latest developments in the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to understand the information being shared
- GS5.** plan and schedule tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures

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- GS7.** take quick decisions to deal with workplaces emergencies/ accidents
- GS8.** analyse work processes to identify improvements
- GS9.** coordinate with the co-workers to achieve the work objectives

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check the availability of required resources</i>	10	12	-	8
PC1. co-ordinate with the relevant company personnel/ third-party supplier to ensure timely delivery of all the necessary apparatus for the installation of BMS such as Direct Digital Controller (DDC), sensors, actuators, relevant types of cables, Human Machine Interface (HMI) display, computer, server, etc.	-	-	-	-
PC2. check that all the components are delivered as per the order	-	-	-	-
PC3. arrange the necessary tools, equipment, and Personal Protective Equipment (PPE) for the installation	-	-	-	-
PC4. ensure the availability of trained personnel to assist with the installation process	-	-	-	-
<i>Check and test the received resources</i>	8	12	-	10
PC5. examine the BMS apparatus to ensure they are not physically damaged	-	-	-	-
PC6. test the BMS apparatus to ensure the correct functioning	-	-	-	-
PC7. report any issues identified with BMS apparatus promptly to the relevant personnel/ supplier	-	-	-	-
PC8. co-ordinate with the relevant personnel/ supplier to get a timely replacement of damaged/ faulty components	-	-	-	-
PC9. maintain the record of receipt/ replacement and testing of the apparatus	-	-	-	-
<i>Prepare for installing the BMS</i>	12	16	-	12
PC10. check that all the civil works are completed in the building and necessary approval/ clearance is obtained before starting the installation process	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. prepare the circuit diagram/ shop drawing/ as-built drawings for the installation of BMS in the building	-	-	-	-
PC12. check that all the outlets related to BMS are accessible and not covered by Mechanical, Electrical and Plumbing (MEP) services	-	-	-	-
PC13. ensure the work area is ready and safe to start the installation of BMS systems	-	-	-	-
PC14. plan the installation process to ensure timely completion as per the client requirements	-	-	-	-
PC15. co-ordinate with the Heating, Ventilation and Air-conditioning (HVAC) installation technician for the installation of HVAC equipment as per the BMS installation plan and client requirements	-	-	-	-
PC16. identify the locations for the installation of various BMS equipment/ devices such as access control, video surveillance, fire alarms, HVAC control, programmable lighting and electric power management as per the BMS installation plan	-	-	-	-
PC17. prepare various equipment for installation as per the manufacturer's instructions	-	-	-	-
NOS Total	30	40	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7205
NOS Name	Prepare for installing the BMS
Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service-I&A
NSQF Level	5
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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ELE/N7206: Carry out the installation of BMS

Description

This NOS unit is about install, connect, and configure Building Management System (BMS) components including conduits, controllers, sensors, peripherals, and actuators as per approved designs, manufacturer guidelines, and system requirements to ensure functional and integrated building automation.

Scope

The scope covers the following :

- Install conduits and carry out cabling
- Install the Direct Digital Controller with field devices
- Install the central peripherals
- Install the motion sensors
- Install the duct air temperature sensor and duct temperature/ humidity sensor
- Install the water differential pressure sensor
- Install the air differential pressure sensor
- Install the immersion water temperature sensor
- Install the smoke detectors in AC ducts
- Install the butterfly valves

Elements and Performance Criteria

Install conduits and carry out wiring

To be competent, the user/individual on the job must be able to:

- PC1.** install Polyvinyl Chloride (PVC)/ Galvanised Iron (GI) conduits between the field devices and the control panel as per the plan
- PC2.** carry out wiring to connect all the field devices as per the approved BMS point schedule, wiring schedule and schematic diagrams

Install the Direct Digital Controller (DDC) with field devices

To be competent, the user/individual on the job must be able to:

- PC3.** install the control panel with the required power supply at the selected location as per the approved shop drawings
- PC4.** install and terminate the temperature sensor, humidity sensor, motorized damper actuators for HVAC equipment as per the manufacturer's instructions
- PC5.** install and terminate the CHW temperature sensor, pressure sensor and flow sensors for CHW pipes as per the manufacturer's instructions
- PC6.** connect all the field devices as per the approved BMS point schedule, wiring schedule and schematic diagrams
- PC7.** install and connect the field devices such as outside air humidity sensor, water level sensor, staircase pressurisation sensor as per the approved BMS system
- PC8.** install the VFD's inside AHU control panels and make the required cable connections as per the approved schematic diagrams

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- PC9.** install the plumbing system including the water booster pump sets, fire pump sets, water calorifier and pump from the DDC panel, as per the system requirements
- PC10.** carry out terminations at field ends as per the manufacturer's recommendations
- PC11.** carry out control cabling for DDC control panels through GI conduit or GI trunking above the false ceiling as per approved shop drawings, schematic drawings
- PC12.** label all the control wires with related identifications and connect as per the approved data point schedule

Install the central peripherals

To be competent, the user/individual on the job must be able to:

- PC13.** carry out bus loop wiring from the Central Controller to all the Variable Air Volume (VAVs) and DDC as per the approved schematic drawings
- PC14.** install the pre-assembled peripherals with controllers hubs in the BMS control room
- PC15.** carry out wiring between the central peripherals and the low current systems as per the approved shop drawings
- PC16.** install the central peripherals such as the computer, printer, monitor as per the approved BMS system and shop drawings

Install the motion sensors

To be competent, the user/individual on the job must be able to:

- PC17.** identify the locations for installing motion sensors as per the BMS plan
- PC18.** set up the mounting brackets at the identified locations
- PC19.** install the motion sensors as per the manufacturer's instructions
- PC20.** connect the motion sensors for BMS digital control

Install the duct air temperature sensor and duct temperature/ humidity sensor

To be competent, the user/individual on the job must be able to:

- PC21.** identify the location of the duct air temperature and duct temperature/ humidity sensor as per the approved shop drawings
- PC22.** drill holes of the recommended diameter using a hole saw cutter and set up the mounting flanges
- PC23.** insert and install the duct air temperature sensor and duct temperature/ humidity sensor on the mounting flanges
- PC24.** ensure the duct temperature/ humidity sensor is mounted on the middle of the duct wall and the sensing element is diagonally bent across the cross-section of the duct
- PC25.** set up the end of the sensing element using the collar provided with the sensor
- PC26.** ensure that the sensing element does not touch the duct wall

Install the water differential pressure sensor

To be competent, the user/individual on the job must be able to:

- PC27.** install the water differential pressure sensor below the pressure measuring points using the GI mounting bracket provided with the sensor
- PC28.** ensure the surface of the installation of the sensor is vibration-free i.e. not on the surface of any equipment with moving parts.
- PC29.** carry out drilling of holes of the recommended diameter at the bottom of the pipes for pressure tapping
- PC30.** use the standard fittings supplied with the sensor to make connections

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PC31. connect the tapping point to the sensor using the copper capillary tube supplied with the sensor

Install the air differential pressure sensor

To be competent, the user/individual on the job must be able to:

PC32. mount the air differential pressure sensor on the duct wall using fasteners supplied with the sensor

PC33. install the sensors using GI mounting bracket provided with the sensor

PC34. install duct probes in the suction and discharge side of the fans

PC35. connect the probes to the pressure sensor using the plastic tubing provided with the sensor

Install the immersion water temperature sensor

To be competent, the user/individual on the job must be able to:

PC36. install the thermowell in the chilled water pipeline and temperature sensor inside the thermowell

PC37. install the immersion water temperature sensor as per the approved shop drawings and manufacturer's recommendations.

PC38. ensure the stem length is completely immersed in chilled water and the cable entry to the sensor is not from the top of the sensor

PC39. apply seal on the socket and sensor threads using Teflon tape

Install the smoke detectors in AC ducts

To be competent, the user/individual on the job must be able to:

PC40. drill mounting holes as per the manufacturer's recommendations for duct housing and to insert air sampling and exhaust tubes

PC41. carry out the wiring for detector connection as per the approved shop drawings

PC42. install the smoke detector with the base unit inside the smoke chamber using the adopter plates

Install the butterfly valves

To be competent, the user/individual on the job must be able to:

PC43. install the butterfly valves in a sandwiched position using appropriate flanges

PC44. install actuators after installing and insulating the valves

PC45. ensure the valve actuators are not installed upside down

PC46. maintain the other actuators in the position recommended by the manufacturer

PC47. connect the two-port valves to the piping using threaded fittings or flanges depending upon the pipe diameter

PC48. erawetrre

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the process of installing (PVC)/ (GI) conduits and carry out cabling to connect all the BMS equipment/ devices

KU2. how to install and terminate the temperature sensor, humidity sensor, motorized damper actuators for HVAC equipment

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- KU3.** how to install and terminate the CHW temperature sensor, pressure sensor and flow sensors for CHW pipes
- KU4.** the process of connecting all the field devices as per the approved BMS point schedule, wiring schedule and schematic diagrams
- KU5.** how to install the field devices such as outside air humidity sensor, water level sensor, staircase pressurisation sensor as per the approved BMS system
- KU6.** the process of installing the VFDs inside AHU control panels as per the approved schematic diagrams
- KU7.** the process of installing conduits and carrying out the wiring for plumbing systems such as water booster pump sets, fire pump sets, water calorifier and pump from the DDC panel as per the system requirements
- KU8.** the process of carrying out terminations at field ends and control cabling for DDC control panels as per approved shop drawings, schematic drawings
- KU9.** the process of installing the central peripherals such as the computer, printer, monitor as per the approved BMS system and shop drawings
- KU10.** the process of installing the motion sensors and connecting them for BMS digital control
- KU11.** how to install the duct air temperature sensor and duct temperature/ humidity sensor
- KU12.** how to install the water differential pressure sensor
- KU13.** the process of installing an air differential pressure sensor
- KU14.** the process of installing an immersion water temperature sensor
- KU15.** how to install smoke detectors in AC ducts
- KU16.** the process of installing butterfly and two-port valves

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to understand the information being shared
- GS5.** plan and schedule tasks for efficient time management
- GS6.** take quick decisions to deal with workplace emergencies/ accident
- GS7.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install conduits and carry out wiring</i>	4	4	-	4
PC1. install Polyvinyl Chloride (PVC)/ Galvanised Iron (GI) conduits between the field devices and the control panel as per the plan	-	-	-	-
PC2. carry out wiring to connect all the field devices as per the approved BMS point schedule, wiring schedule and schematic diagrams	-	-	-	-
<i>Install the Direct Digital Controller (DDC) with field devices</i>	4	6	-	4
PC3. install the control panel with the required power supply at the selected location as per the approved shop drawings	-	-	-	-
PC4. install and terminate the temperature sensor, humidity sensor, motorized damper actuators for HVAC equipment as per the manufacturer's instructions	-	-	-	-
PC5. install and terminate the CHW temperature sensor, pressure sensor and flow sensors for CHW pipes as per the manufacturer's instructions	-	-	-	-
PC6. connect all the field devices as per the approved BMS point schedule, wiring schedule and schematic diagrams	-	-	-	-
PC7. install and connect the field devices such as outside air humidity sensor, water level sensor, staircase pressurisation sensor as per the approved BMS system	-	-	-	-
PC8. install the VFD's inside AHU control panels and make the required cable connections as per the approved schematic diagrams	-	-	-	-
PC9. install the plumbing system including the water booster pump sets, fire pump sets, water calorifier and pump from the DDC panel, as per the system requirements	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. carry out terminations at field ends as per the manufacturer's recommendations	-	-	-	-
PC11. carry out control cabling for DDC control panels through GI conduit or GI trunking above the false ceiling as per approved shop drawings, schematic drawings	-	-	-	-
PC12. label all the control wires with related identifications and connect as per the approved data point schedule	-	-	-	-
<i>Install the central peripherals</i>	4	4	-	4
PC13. carry out bus loop wiring from the Central Controller to all the Variable Air Volume (VAVs) and DDC as per the approved schematic drawings	-	-	-	-
PC14. install the pre-assembled peripherals with controllers hubs in the BMS control room	-	-	-	-
PC15. carry out wiring between the central peripherals and the low current systems as per the approved shop drawings	-	-	-	-
PC16. install the central peripherals such as the computer, printer, monitor as per the approved BMS system and shop drawings	-	-	-	-
<i>Install the motion sensors</i>	2	2	-	4
PC17. identify the locations for installing motion sensors as per the BMS plan	-	-	-	-
PC18. set up the mounting brackets at the identified locations	-	-	-	-
PC19. install the motion sensors as per the manufacturer's instructions	-	-	-	-
PC20. connect the motion sensors for BMS digital control	-	-	-	-
<i>Install the duct air temperature sensor and duct temperature/ humidity sensor</i>	4	6	-	4

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. identify the location of the duct air temperature and duct temperature/ humidity sensor as per the approved shop drawings	-	-	-	-
PC22. drill holes of the recommended diameter using a hole saw cutter and set up the mounting flanges	-	-	-	-
PC23. insert and install the duct air temperature sensor and duct temperature/ humidity sensor on the mounting flanges	-	-	-	-
PC24. ensure the duct temperature/ humidity sensor is mounted on the middle of the duct wall and the sensing element is diagonally bent across the cross-section of the duct	-	-	-	-
PC25. set up the end of the sensing element using the collar provided with the sensor	-	-	-	-
PC26. ensure that the sensing element does not touch the duct wall	-	-	-	-
<i>Install the water differential pressure sensor</i>	2	4	-	4
PC27. install the water differential pressure sensor below the pressure measuring points using the GI mounting bracket provided with the sensor	-	-	-	-
PC28. ensure the surface of the installation of the sensor is vibration-free i.e. not on the surface of any equipment with moving parts.	-	-	-	-
PC29. carry out drilling of holes of the recommended diameter at the bottom of the pipes for pressure tapping	-	-	-	-
PC30. use the standard fittings supplied with the sensor to make connections	-	-	-	-
PC31. connect the tapping point to the sensor using the copper capillary tube supplied with the sensor	-	-	-	-
<i>Install the air differential pressure sensor</i>	2	2	-	1
PC32. mount the air differential pressure sensor on the duct wall using fasteners supplied with the sensor	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. install the sensors using GI mounting bracket provided with the sensor	-	-	-	-
PC34. install duct probes in the suction and discharge side of the fans	-	-	-	-
PC35. connect the probes to the pressure sensor using the plastic tubing provided with the sensor	-	-	-	-
<i>Install the immersion water temperature sensor</i>	4	4	-	2
PC36. install the thermowell in the chilled water pipeline and temperature sensor inside the thermowell	-	-	-	-
PC37. install the immersion water temperature sensor as per the approved shop drawings and manufacturer's recommendations.	-	-	-	-
PC38. ensure the stem length is completely immersed in chilled water and the cable entry to the sensor is not from the top of the sensor	-	-	-	-
PC39. apply seal on the socket and sensor threads using Teflon tape	-	-	-	-
<i>Install the smoke detectors in AC ducts</i>	2	4	-	1
PC40. drill mounting holes as per the manufacturer's recommendations for duct housing and to insert air sampling and exhaust tubes	-	-	-	-
PC41. carry out the wiring for detector connection as per the approved shop drawings	-	-	-	-
PC42. install the smoke detector with the base unit inside the smoke chamber using the adopter plates	-	-	-	-
<i>Install the butterfly valves</i>	2	4	-	2
PC43. install the butterfly valves in a sandwiched position using appropriate flanges	-	-	-	-
PC44. install actuators after installing and insulating the valves	-	-	-	-
PC45. ensure the valve actuators are not installed upside down	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC46. maintain the other actuators in the position recommended by the manufacturer	-	-	-	-
PC47. connect the two-port valves to the piping using threaded fittings or flanges depending upon the pipe diameter	-	-	-	-
PC48. erawetrre	-	-	-	-
NOS Total	30	40	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7206
NOS Name	Carry out the installation of BMS
Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service-I&A
NSQF Level	5
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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ELE/N7208: Carry out commissioning and testing of BMS

Description

This NOS unit is about to perform pre-commissioning and commissioning of Building Management System (BMS) components, including DDC panels, FCUs, and third-party systems, ensuring proper installation, wiring, functionality, and communication as per approved procedures and standards.

Scope

The scope covers the following :

- Carry out pre-commissioning of BMS
- Carry out commissioning of DDC panels
- Carry out commissioning of the Fan Coil Unit (FCU)
- Test the digital and analogue inputs and outputs
- Test the communication link, printer and alarms
- Test the third-party system interface connectivity

Elements and Performance Criteria

Carry out pre-commissioning of BMS

To be competent, the user/individual on the job must be able to:

- PC1.** ensure that the installation of all BMS components is complete, as per with the approved shop drawings and manufacturer's recommendations
- PC2.** check that mechanical and electrical installation inspection recommendations are incorporated, ensuring no damage during mechanical completion and pre-commissioning
- PC3.** replace any damaged components as per the SOP
- PC4.** ensure all the relevant systems are pre-commissioned and commissioned as standalone systems
- PC5.** check all peripheral devices are mounted and connected
- PC6.** check all mechanical installations for the correct location and application
- PC7.** check all the cables are connected correctly inside the control panels/ DDC controllers and to the peripheral devices
- PC8.** check the control panel input terminals for interference voltages using an AC range voltmeter
- PC9.** ensure all Fan Coil Unit (FCU) controllers are installed and connected via communication bus
- PC10.** ensure the installation of all the sensors, DP switches, valves is completed both mechanically and electrically
- PC11.** test all control cables point to point using a multimeter

Carry out commissioning of DDC panels

To be competent, the user/individual on the job must be able to:

- PC12.** check the DDC panels have the correct power supply, ensuring personal safety
- PC13.** install the DDC controller software for correct automation of DDC, as per the instructions included with the licensed software

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PC14. set the controller address, date and time and check for any system alarms

PC15. connect the field wiring at DDC controller

Carry out commissioning of the Fan Coil Unit (FCU)

To be competent, the user/individual on the job must be able to:

PC16. check for 240Vac at Miniature Circuit Breaker (MCB) or spur socket as applicable

PC17. check the status of the LEDs for communications after switching on the power supply module and check for any alarm

PC18. check that the network communication link is up between the BMS servers, workstations, DDC Panels, LON to IP converters (LIP) and Building Automation and Control Network (BACnet) controllers

PC19. check the values displayed on the graphics screen and confirm that they are linked to the points of concerned plant as per the DDC I/O testing and commissioning check list

PC20. test that changes occur in field as per the sequence of operation and the changes are displayed correctly on the graphics screen

Test the digital and analogue inputs and outputs

To be competent, the user/individual on the job must be able to:

PC21. check all the Digital Inputs (DI) and Digital Outputs (DO) cables coming from field are volt free

PC22. connect each pair of field instrument cable at the appropriate terminals as per the panel wiring termination drawing

PC23. shorten and disconnect the wires at the field end and check for appropriate LED status at the DI module

PC24. change the Hand-Off-Auto selector switches for all the equipment to auto position

PC25. connect all cables of field instruments for analogue inputs/ outputs at the appropriate terminals as per the panel wiring termination drawing

PC26. check for availability of 24Vac at each valve motor and field instrument

PC27. connect all cables of field instruments for analogue inputs/ outputs at the appropriate terminals as per the panel wiring termination drawing

Test the communication link, printer and alarms

To be competent, the user/individual on the job must be able to:

PC28. check that the communication link is up between the servers/ workstations and printers

PC29. check the drivers are correctly installed as per the manufacturer's instructions

PC30. test each alarm in the system and validate that the system generates the appropriate alarm message, the message appears at workstations and printers, and any other related actions occur as defined

Test the third-party system interface connectivity

To be competent, the user/individual on the job must be able to:

PC31. match the values on the BMS graphics and on the third party system display for different parameters in real time after the third party systems are configured

PC32. check and verify the sequence of operations according to the Sequence of Operation Manual after all field cables are verified and checked

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the process of pre-commissioning a BMS
- KU2.** the importance of ensuring that all peripheral devices are mounted and connected
- KU3.** the importance of checking the mechanical installations for the correct location and application
- KU4.** how to check if all the cables are connected correctly inside the control panels/ DDC controllers and to the peripheral devices
- KU5.** the process of checking the control panel input terminals for interference voltages with the use of an AC range voltmeter
- KU6.** how to check if the Fan Coil Unit (FCU) controllers are installed and connected via communication bus
- KU7.** the importance of ensuring the installation of all the sensors, DP switches, valves is completed both mechanically and electrically
- KU8.** use of a multimeter to test all control cables
- KU9.** the process of commissioning DDC panels including the installation of relevant DDC controller application software
- KU10.** the process of testing the digital input/ output and analogue input/ output
- KU11.** the process of commissioning the Fan Coil Unit (FCU)
- KU12.** the importance of checking the network communication between the BMS servers, workstations, DDC Panels, LON to IP converters (LIP) and Building Automation and Control Network (BACnet) controllers
- KU13.** the process of testing the sequence of operation according to changes in the field devices
- KU14.** the process of checking the printer and alarms for correct functioning
- KU15.** how to check that the communication link is up between the servers/ workstations and printers
- KU16.** how to check the third-party system interface connectivity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to understand the information being shared
- GS5.** plan and schedule tasks for efficient time management
- GS6.** take quick decisions to deal with workplace emergencies/ accident
- GS7.** evaluate all possible solutions to a problem to select the best one

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out pre-commissioning of BMS</i>	5	10	-	5
PC1. ensure that the installation of all BMS components is complete, as per with the approved shop drawings and manufacturer's recommendations	-	-	-	-
PC2. check that mechanical and electrical installation inspection recommendations are incorporated, ensuring no damage during mechanical completion and pre-commissioning	-	-	-	-
PC3. replace any damaged components as per the SOP	-	-	-	-
PC4. ensure all the relevant systems are pre-commissioned and commissioned as standalone systems	-	-	-	-
PC5. check all peripheral devices are mounted and connected	-	-	-	-
PC6. check all mechanical installations for the correct location and application	-	-	-	-
PC7. check all the cables are connected correctly inside the control panels/ DDC controllers and to the peripheral devices	-	-	-	-
PC8. check the control panel input terminals for interference voltages using an AC range voltmeter	-	-	-	-
PC9. ensure all Fan Coil Unit (FCU) controllers are installed and connected via communication bus	-	-	-	-
PC10. ensure the installation of all the sensors, DP switches, valves is completed both mechanically and electrically	-	-	-	-
PC11. test all control cables point to point using a multimeter	-	-	-	-
<i>Carry out commissioning of DDC panels</i>	5	5	-	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check the DDC panels have the correct power supply, ensuring personal safety	-	-	-	-
PC13. install the DDC controller software for correct automation of DDC, as per the instructions included with the licensed software	-	-	-	-
PC14. set the controller address, date and time and check for any system alarms	-	-	-	-
PC15. connect the field wiring at DDC controller	-	-	-	-
<i>Carry out commissioning of the Fan Coil Unit (FCU)</i>	5	5	-	5
PC16. check for 240Vac at Miniature Circuit Breaker (MCB) or spur socket as applicable	-	-	-	-
PC17. check the status of the LEDs for communications after switching on the power supply module and check for any alarm	-	-	-	-
PC18. check that the network communication link is up between the BMS servers, workstations, DDC Panels, LON to IP converters (LIP) and Building Automation and Control Network (BACnet) controllers	-	-	-	-
PC19. check the values displayed on the graphics screen and confirm that they are linked to the points of concerned plant as per the DDC I/O testing and commissioning check list	-	-	-	-
PC20. test that changes occur in field as per the sequence of operation and the changes are displayed correctly on the graphics screen	-	-	-	-
<i>Test the digital and analogue inputs and outputs</i>	5	10	-	5
PC21. check all the Digital Inputs (DI) and Digital Outputs (DO) cables coming from field are volt free	-	-	-	-
PC22. connect each pair of field instrument cable at the appropriate terminals as per the panel wiring termination drawing	-	-	-	-
PC23. shorten and disconnect the wires at the field end and check for appropriate LED status at the DI module	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. change the Hand-Off-Auto selector switches for all the equipment to auto position	-	-	-	-
PC25. connect all cables of field instruments for analogue inputs/ outputs at the appropriate terminals as per the panel wiring termination drawing	-	-	-	-
PC26. check for availability of 24Vac at each valve motor and field instrument	-	-	-	-
PC27. connect all cables of field instruments for analogue inputs/ outputs at the appropriate terminals as per the panel wiring termination drawing	-	-	-	-
<i>Test the communication link, printer and alarms</i>	5	5	-	5
PC28. check that the communication link is up between the servers/ workstations and printers	-	-	-	-
PC29. check the drivers are correctly installed as per the manufacturer's instructions	-	-	-	-
PC30. test each alarm in the system and validate that the system generates the appropriate alarm message, the message appears at workstations and printers, and any other related actions occur as defined	-	-	-	-
<i>Test the third-party system interface connectivity</i>	5	5	-	5
PC31. match the values on the BMS graphics and on the third party system display for different parameters in real time after the third party systems are configured	-	-	-	-
PC32. check and verify the sequence of operations according to the Sequence of Operation Manual after all field cables are verified and checked	-	-	-	-
NOS Total	30	40	-	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7208
NOS Name	Carry out commissioning and testing of BMS
Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service-I&A
NSQF Level	5
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

ELE/N7207: Carry out repair and maintenance of BMS

Description

This NOS unit is about perform repair, maintenance, and record-keeping of BMS field devices and electrical panels, manage helpdesk operations, address complaints, and ensure proper system functionality and documentation as per standard procedures.

Scope

The scope covers the following :

- Carry out repair and maintenance of BMS field devices
- Carry out repair and maintenance of electrical panels
- Manage the BMS helpdesk, complaints and requests
- Maintain the records

Elements and Performance Criteria

Carry out repair and maintenance of BMS field devices

To be competent, the user/individual on the job must be able to:

- PC1.** test the valve actuators, damper actuator, sensors, transducers, high-low limit thermostats, frost thermostats, pressure switches, control valve, relays, emergency stop button and fire/smoke shut-down, the shut-off operation of spring return actuators and Digital Input and Output (I/O) for the correct functioning
- PC2.** check the field devices for wear and tear or damage
- PC3.** check the BMS is connected with the control system and is online
- PC4.** check repair and maintenance needs of the BMS control system
- PC5.** check the motion sensors are working as expected
- PC6.** clean the valve actuators, damper actuators and sensors
- PC7.** re-calibrate all sensors, transducers, valve start points and actuator travel times as per the requirement
- PC8.** replace the worn out or damaged field devices as per the Standard Operating Procedure
- PC9.** carry out maintenance of the motion sensors or replace them as required
- PC10.** carry out control panel modifications, up-gradation and installations as required

Carry out repair and maintenance of electrical panels

To be competent, the user/individual on the job must be able to:

- PC11.** examine the panel exterior for signs of damage
- PC12.** check the tightness of all the electrical terminations
- PC13.** check the integrity of power cabling insulation
- PC14.** check settings, ratings and operation of protective devices such as overloads, residual current devices, circuit breakers and fuses
- PC15.** check for signs of overheating of components such as contactors, cables, connectors, as well as the internal temperature within the panel

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- PC16.** check and re-torque the busbar nuts and bolts
- PC17.** inspect the transformers and power supplies for wear and tear or damage
- PC18.** check the panel is correctly earth-bonded
- PC19.** ensure main door electrical isolator is engaged and operational
- PC20.** ensure ventilation fans and grilles are functional and clean
- PC21.** ensure the isolators, relays, contactors and starters are functioning correctly and free from pitting
- PC22.** ensure the incoming power supply voltages are within the prescribed limits
- PC23.** ensure the cleanliness of panel and components and risk of ingress of dirt or moisture is minimised
- PC24.** ensure status indicators on all panels are functional

Manage the BMS helpdesk, complaints and requests

To be competent, the user/individual on the job must be able to:

- PC25.** manage the BMS help desk, providing a central point for all the BMS-related requests/ complaints
- PC26.** ensure closure of BMS-related requests/ complaints with timely resolution
- PC27.** direct the BMS complaints requiring vendor attention to the concerned vendor promptly
- PC28.** escalate the unresolved complaints to the relevant authority as per the escalation matrix

Maintain the records

To be competent, the user/individual on the job must be able to:

- PC29.** maintain the logs of calls reporting any issues with the BMS
- PC30.** maintain and update documents related to BMS room, device hosting and its maintenance
- PC31.** maintain records and logs regarding the building temperatures, energy consumption, control panel readings, issues encountered and the steps taken to correct the problem, etc.
- PC32.** prepare the daily/ weekly/ monthly Management Information Systems (MIS) report regarding the requests/ complaints received and share with the relevant authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the process of testing the valve actuators, damper actuator, sensors, transducers, high-low limit thermostats, frost thermostats, pressure switches, control valve, relays, emergency stop button and fire/ smoke shut-down, the shut-off operation of spring return actuators and Digital Input and Output (I/O) for the correct functioning
- KU2.** the importance of ensuring the field devices are not worn-out or damaged
- KU3.** how to check the BMS is connected with the control system and is online
- KU4.** how to identify the repair and maintenance needs of the BMS control system
- KU5.** how to check if the motion sensors are working as expected and replacing them if required
- KU6.** the process of cleaning the valve actuators, damper actuators and sensors
- KU7.** the process of re-calibrating all sensors, transducers, valve start points and actuator travel times
- KU8.** the process of replacing the worn out or damaged field devices

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- KU9.** how to check settings, ratings and operation of protective devices such as overloads, residual current devices, circuit breakers and fuses
- KU10.** signs of overheating of components such as contactors, cables, connectors and internal temperature within the electrical panels
- KU11.** the process of re-torquing the busbar nuts and bolts
- KU12.** how to check if the electrical panel is earth-bonded correctly
- KU13.** how to check if the main door electrical isolator is engaged and operational
- KU14.** the importance of ensuring ventilation fans and grilles are functional and clean
- KU15.** the importance of ensuring isolators, relays, contactors and starters are functioning correctly and free from pitting
- KU16.** the importance of ensuring incoming power supply voltages are within the prescribed limits
- KU17.** the importance of ensuring status indicators on all panels are functional
- KU18.** the importance of ensuring the closure of BMS-related requests/ complaints with timely resolution
- KU19.** the importance of directing the BMS complaints requiring vendor attention to the concerned vendor promptly and escalating the unresolved complaints to the relevant authority as per the escalation matrix
- KU20.** various records to be maintained regarding the BMS operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make work-related records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to comprehend the information being shared
- GS5.** take quick decisions to resolve work-related issues to minimise the impact on productivity
- GS6.** plan and prioritise tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out repair and maintenance of BMS field devices</i>	8	12	-	8
PC1. test the valve actuators, damper actuator, sensors, transducers, high-low limit thermostats, frost thermostats, pressure switches, control valve, relays, emergency stop button and fire/ smoke shut-down, the shut-off operation of spring return actuators and Digital Input and Output (I/O) for the correct functioning	-	-	-	-
PC2. check the field devices for wear and tear or damage	-	-	-	-
PC3. check the BMS is connected with the control system and is online	-	-	-	-
PC4. check repair and maintenance needs of the BMS control system	-	-	-	-
PC5. check the motion sensors are working as expected	-	-	-	-
PC6. clean the valve actuators, damper actuators and sensors	-	-	-	-
PC7. re-calibrate all sensors, transducers, valve start points and actuator travel times as per the requirement	-	-	-	-
PC8. replace the worn out or damaged field devices as per the Standard Operating Procedure	-	-	-	-
PC9. carry out maintenance of the motion sensors or replace them as required	-	-	-	-
PC10. carry out control panel modifications, up-gradation and installations as required	-	-	-	-
<i>Carry out repair and maintenance of electrical panels</i>	10	10	-	10
PC11. examine the panel exterior for signs of damage	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check the tightness of all the electrical terminations	-	-	-	-
PC13. check the integrity of power cabling insulation	-	-	-	-
PC14. check settings, ratings and operation of protective devices such as overloads, residual current devices, circuit breakers and fuses	-	-	-	-
PC15. check for signs of overheating of components such as contactors, cables, connectors, as well as the internal temperature within the panel	-	-	-	-
PC16. check and re-torque the busbar nuts and bolts	-	-	-	-
PC17. inspect the transformers and power supplies for wear and tear or damage	-	-	-	-
PC18. check the panel is correctly earth-bonded	-	-	-	-
PC19. ensure main door electrical isolator is engaged and operational	-	-	-	-
PC20. ensure ventilation fans and grilles are functional and clean	-	-	-	-
PC21. ensure the isolators, relays, contactors and starters are functioning correctly and free from pitting	-	-	-	-
PC22. ensure the incoming power supply voltages are within the prescribed limits	-	-	-	-
PC23. ensure the cleanliness of panel and components and risk of ingress of dirt or moisture is minimised	-	-	-	-
PC24. ensure status indicators on all panels are functional	-	-	-	-
<i>Manage the BMS helpdesk, complaints and requests</i>	8	12	-	8
PC25. manage the BMS help desk, providing a central point for all the BMS-related requests/ complaints	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. ensure closure of BMS-related requests/ complaints with timely resolution	-	-	-	-
PC27. direct the BMS complaints requiring vendor attention to the concerned vendor promptly	-	-	-	-
PC28. escalate the unresolved complaints to the relevant authority as per the escalation matrix	-	-	-	-
<i>Maintain the records</i>	4	6	-	4
PC29. maintain the logs of calls reporting any issues with the BMS	-	-	-	-
PC30. maintain and update documents related to BMS room, device hosting and its maintenance	-	-	-	-
PC31. maintain records and logs regarding the building temperatures, energy consumption, control panel readings, issues encountered and the steps taken to correct the problem, etc.	-	-	-	-
PC32. prepare the daily/ weekly/ monthly Management Information Systems (MIS) report regarding the requests/ complaints received and share with the relevant authority	-	-	-	-
NOS Total	30	40	-	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N7207
NOS Name	Carry out repair and maintenance of BMS
Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service-I&A
NSQF Level	5
Credits	5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N7205.Prepare for installing the BMS	30	40	-	30	100	20
ELE/N7206.Carry out the installation of BMS	30	40	-	30	100	20
ELE/N7208.Carry out commissioning and testing of BMS	30	40	-	30	100	20
ELE/N7207.Carry out repair and maintenance of BMS	30	40	-	30	100	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	20
Total	140	190	-	120	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.